Singular Gaussian Conditioning via the Inverse-Free Schur Complement

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Abstract

Singular Gaussian conditioning in Hilbert space is governed by Anderson–Trapp shorting, while the finite-dimensional Schur formula $A - BD^{-1}B^*$ is only its invertible-block shadow. In trace-class Hilbert problems, compact covariance blocks lack bounded ambient inverses, so shorting replaces direct inversion through a variational principle on the Cameron–Martin range. The rank-one canonical scaling is $\Sigma_{\lambda,kk} = \varepsilon_0/(2\lambda^2)$ and the shorting subtraction gives $S_{k,\lambda} = \frac{\varepsilon_0}{2\lambda^2} - R_\lambda$ with $R_\lambda = O(\lambda^{-6})$. For Gaussian Radon laws, this yields the finite-part functional expansion

$$\mathfrak{J}_\lambda(\mu_{\text{obs}}; \mu_\lambda) = \frac{b_k^2}{\varepsilon_0} + O(\lambda^{-4}),$$

with a kernel-independent limit at rank one.

Keywords: Block operators; Anderson–Trapp shorting; Schur complementation; Cameron–Martin range; trace-class covariance; Fredholm finite parts; Gaussian disintegration; Galerkin approximation

1. Introduction

The starting point is a functional-analytic identification, not a probabilistic regularization. In finite dimensions, the Schur complement of a positive block matrix is written as $A-BD^{-1}B^*$. This formula is not the primitive operation: it is the invertible-block coordinate expression of Anderson–Trapp shorting. For covariance operators on a separable Hilbert space, the free block D is typically compact and trace class; it therefore has no bounded inverse on the ambient space [1–4]. Singular conditioning therefore does not repair the Schur formula. It exposes that Schur complementation was the finite-dimensional shadow of shorting all along.

A systematic impossibility result (Theorem 9) shows that direct inversion fails at three independent levels: algebraically there is no bounded inverse, numerically Galerkin truncations suffer spectral ill-conditioning $\kappa \sim N^2$, and asymptotically the regime may require $\lambda \gg N^2$, placing the true limit beyond the reach of any fixed truncation. Classical Schur-complement and shorted-operator theory, originating in Krein's theory of non-negative extensions and developed by Anderson and Trapp, gives the underlying variational formula for positive block operators [5–7]. Recent Schur-complement work on non-negative operators and complementable block operators gives a current operator-theoretic context for the same mechanism [8, 9]. In the covariance setting, its admissibility condition is exactly a Cameron–Martin range condition: cross-covariance vectors must have finite covariance energy rather than lie in the domain of a non-existent bounded inverse. This paper uses that range condition to define the shorted residual as an energy projection, proves its λ^{-6} scale, and shows that Galerkin approximation is Rayleigh–Ritz restriction of the same supremum to nested finite-dimensional test spaces.

Gaussian conditioning enters after this operator identification is in place. Singular coordinate conditioning reaches the Feldman–Hájek measure-class boundary: exact evidence is supported on a hyperplane of prior measure zero, while positive-noise Gaussian inverse-problem formulas may remain in the equivalent-measure regime [1, 10–14]. The shorted covariance separates the finite residual from the diverging normalization terms. Fredholm determinant identities and determinant-and-trace cancellation then extract a canonical finite part.

Thus the paper's logical order is: shorting first, Gaussian disintegration second. The stochastic terminology is an application layer. The structural object is Anderson–Trapp shorting; the familiar Schur formula is its finite-dimensional invertible-block expression.

In the rank-one case, the residual has the exact asymptotic form

$$S_{k,\lambda} = \frac{\varepsilon_0}{2\lambda^2} - R_\lambda, \qquad R_\lambda = O(\lambda^{-6}),$$

hence

$$S_{k,\lambda}^{-1} = \frac{2}{\varepsilon_0}\lambda^2 + O(\lambda^{-2}).$$

These are the powers used in the finite-part cancellation: the target block contributes the λ^2 pole, while the shorted residual is too small to alter the finite part.

The paper has four main contributions:

1. **Block Lyapunov normalization.** We identify the operator mechanism that fixes the target block at the unique finite-part scale $\Sigma_{\lambda,kk} = \varepsilon_0/(2\lambda^2)$.

2. **Schur as shorting.** We identify the formal inverse expression $A - BD^{-1}B^*$ as the invertible-block expression of Anderson–Trapp shorting, define the residual by the variational short on the Cameron–Martin range, and prove the $O(\lambda^{-6})$ shorted-residual estimate.

3. **Galerkin monotonicity.** We prove that finite-rank Galerkin truncations converge monotonically by expanding the variational supremum. Loewner monotonicity is a consequence of nested subspaces, not matrix inversion.

4. **Fredholm finite part.** For Gaussian Radon laws, disintegration supplies the exact-evidence measure, while Weinstein–Aronszajn and Fredholm determinant identities cancel the trace anomalies and reduce the finite part to the invariant quadratic form.

1.1. Positioning in the literature

The paper is closest to four bodies of work. First, it uses Schur complements, shorted operators, Douglas range inclusion, and modern complementability theory for block operators [5–9, 15–17]. Second, it uses the measure-class structure of Gaussian measures, especially the Cameron–Martin theorem and Feldman–Hájek dichotomy [1, 10, 11]. Third, it is adjacent to Bayesian inverse problems and Hilbert-space Gaussian conditioning, including shorted covariance descriptions, positive noise inverse-problem formulas, low-rank covariance approximations, and nonlinear observation maps [12–14, 18–21]. Fourth, the examples come from stochastic evolution equations and elliptic Gaussian fields, where dissipative sectorial generators produce compact trace-class stationary covariances naturally approximated by Galerkin truncation [2–4, 22, 23].

1.2. Organization of the paper

Section 2 sets the operator model and canonical normalization. Section 3.4 explains why direct inversion cannot govern the limit. Section 3 gives the Anderson–Trapp shorting construction and its finite-dimensional shadow. Section 4 uses this shorted residual in Gaussian disintegration and the finite-part functional. Section 5 closes with an elliptic benchmark. Section 6 concludes. Appendix proofs are organized as: calibrated block identities and exact-resolvent formulas (Section Appendix A), bounded-inverse Schur shadows (Section Appendix B), Galerkin shorting consistency (Section Appendix C), Gaussian Fredholm analysis (Section Appendix D), and admissibility geometry (Section Appendix E). Numerical diagnostics are collected in the supplementary material and are not proof dependencies.

2. Operator Model and Canonical Normalization

2.1. Operator Evolution and Regularized Conditioning

We model the continuous-time process as stochastic evolution on a separable Hilbert space $\mathcal{H}$ [3]:

$$dX_t = \mathcal{A}X_t\,dt + \mathcal{D}^{1/2}dW_t \tag{1}$$

where $\mathcal{A}$ is the drift operator and $\mathcal{D}$ is the trace-class diffusion operator. We decompose the Hilbert space as $\mathcal{H} = \mathcal{H}_k \oplus \mathcal{H}_I$, where $\mathcal{H}_k := \mathbb{R}\phi_k$ is the one-dimensional target coordinate subspace, and $\mathcal{H}_I := \mathcal{H}_k^{\perp}$ is the free bulk subspace. Let $\Pi_k := \phi_k \otimes \phi_k$ and $\Pi_I := I - \Pi_k$ be the corresponding orthogonal projections.

To approximate the exact point conditioning $X_k = x_k^*$, we set the incoming target-coordinate blocks to zero and apply a regularization parameter $\lambda > 0$.

Definition 1 (Calibrated Approximating Operator Family)**.** The regularized conditioning operator of strength $\lambda > 0$ at coordinate X_k acts on the operator evolution (1) by modifying both the drift and the diffusion:

$$\mathcal{A}^{(\lambda)} := \mathcal{A}_{\text{cond}} - \lambda\Pi_k, \qquad \mathcal{D}^{(\lambda)} := \mathcal{D}_{\text{cond}} + d_\lambda\Pi_k \tag{2}$$

where $\mathcal{A}_{\text{cond}}$ is the block drift operator with incoming target-coordinate blocks set to zero ($\Pi_k\mathcal{A}_{\text{cond}} = 0$), $\mathcal{D}_{\text{cond}}$ has target-coordinate cross-noise set to zero, and $d_\lambda > 0$ is the local diffusion variance of the target coordinate at regularization strength λ. In the SDE, this corresponds to:

$$dX_{k,t} = -\lambda(X_{k,t} - x_k^*)dt + \sqrt{d_\lambda}\,dW_{k,t}. \tag{3}$$

2.2. Canonical Normalization

The block-decoupled structure under the regularized operator family yields:

$$\mathcal{A}^{(\lambda)} = \begin{pmatrix} -\lambda & 0 \\ a_{Ik} & \mathcal{A}_{II} \end{pmatrix}, \qquad \mathcal{D}^{(\lambda)} = \begin{pmatrix} d_\lambda & 0 \\ 0 & D_{II} \end{pmatrix}, \tag{4}$$

where $a_{Ik} \coloneqq \mathcal{A}_{Ik}\phi_k \in \mathcal{H}_I$ represents the off-diagonal drift coupling. Let Σ_λ denote the stationary covariance operator under regularization strength λ, solving the Lyapunov equation:

$$\mathcal{A}^{(\lambda)}\Sigma_\lambda + \Sigma_\lambda(\mathcal{A}^{(\lambda)})^* + \mathcal{D}^{(\lambda)} = 0. \tag{5}$$

The target block kk decouples completely from the free coordinates, yielding the exact scalar equation:

$$-2\lambda\Sigma_{\lambda,kk} + d_\lambda = 0, \qquad \Sigma_{\lambda,kk} = \frac{d_\lambda}{2\lambda}. \tag{6}$$

The normalization is fixed by canonical scaling, not by a free modeling convention. In the Gaussian finite-part extension below, a constant forcing b_k produces the deterministic target offset

$$\Delta_\lambda \coloneqq m_{\lambda,k} - x_k^* = \frac{b_k}{\lambda} = O(\lambda^{-1}), \tag{7}$$

so the squared deterministic displacement has the rigid scale $\Delta_\lambda^2 = b_k^2/\lambda^2 = O(\lambda^{-2})$. The leading target contribution to the conditional finite-part functional is the signal-to-noise ratio

$$\frac{\Delta_\lambda^2}{2\Sigma_{\lambda,kk}} = \frac{b_k^2/\lambda^2}{2\Sigma_{\lambda,kk}}. \tag{8}$$

Consequently the target covariance must decay at the same order as the squared deterministic displacement. If $\Sigma_{\lambda,kk} = O(\lambda^{-1})$, as for the slow family $d_\lambda = \varepsilon_0$, then (8) is $O(\lambda^{-1})$ and the finite target contribution collapses to zero. If $\Sigma_{\lambda,kk} = O(\lambda^{-3})$, as for the fast family $d_\lambda = \varepsilon_0/\lambda^2$, then (8) is $O(\lambda)$ and the target contribution diverges. The numerical supplement reports these slow and fast scaling diagnostics, but the conclusion already follows from the exact scalar identity. A nonzero finite limit therefore forces the critical covariance scale

$$\Sigma_{\lambda,kk} \asymp \lambda^{-2}. \tag{9}$$

Proposition 2 (Canonical Scaling)**.** *The unique target-diffusion scaling that makes the deterministic offset and target variance balance at a finite nonzero scale is*

$$\Sigma_{\lambda,kk} \equiv \frac{\varepsilon_0}{2\lambda^2}, \qquad \varepsilon_0 > 0. \tag{10}$$

By the scalar Lyapunov identity (6)*, this is equivalent to*

$$d_\lambda = 2\lambda\Sigma_{\lambda,kk} = \frac{\varepsilon_0}{\lambda} \quad \textit{for every } \lambda > 0. \tag{11}$$

With this canonical family,

$$\frac{\Delta_\lambda^2}{2\Sigma_{\lambda,kk}} = \frac{b_k^2}{\varepsilon_0}, \tag{12}$$

and the calibrated approximating operators become

$$dX_{k,t} = -\lambda(X_{k,t} - x_k^*)dt + \sqrt{\varepsilon_0/\lambda}\, dW_{k,t}. \tag{13}$$

All subsequent results are stated for this canonical scaling family.

Definition 3 (Shorted Schur Energy Invariant)**.** Let $S_{k,\lambda} := \|X_k - P_{\mathcal{M}_I} X_k\|_{L^2}^2$ represent the shorted Schur projection residual of the target coordinate X_k against the closed linear span $\mathcal{M}_I$ of the free coordinates X_I. In probabilistic language this is the linear minimum mean squared error residual, but the operator object is the Schur energy projection. The *shorted Schur invariant* m_2^{Schur} is defined as:

$$m_2^{\text{Schur}} := \lim_{\lambda\to\infty} \lambda^2 S_{k,\lambda}. \tag{14}$$

2.3. Operator Setup and Assumptions

We use one core assumption package for the shorting limit:

Assumption 4 (Core Analytical Assumptions for Singular Hyperplane Conditioning)**.** (i) **Trace-Class Lyapunov framework:** For each $\lambda > 0$, the stationary covariance Σ_λ is the unique strictly positive solution to the trace-class Lyapunov equation

$$\mathcal{A}^{(\lambda)}\Sigma_\lambda + \Sigma_\lambda \mathcal{A}^{(\lambda)*} + \mathcal{D}^{(\lambda)} = 0. \tag{15}$$

The free-block covariance under full decoupling is $Q_0 := \Sigma_{II}^{(0)}$, assumed compact, injective, trace-class, and positive.

(ii) **Cameron–Martin admissibility of coupling:** The block-decupled coefficients satisfy $\mathcal{A}_{kI} = \mathcal{D}_{kI} = \mathcal{D}_{Ik} = 0$ and $\mathcal{A}_{kk} = -\lambda$. Let $H_{Q_0} = \mathrm{Range}(Q_0^{1/2})$ and $|u|_{Q_0} = \|Q_0^{-1/2}u\|_{\mathcal{H}_I}$. The leakage vector $a_{Ik} = \mathcal{A}_{Ik}\phi_k$ belongs to H_{Q_0}, and on H_{Q_0} the restricted drift semigroup is exponentially stable:

$$\|e^{t\mathcal{A}_{II}}\|_{\mathcal{L}(H_{Q_0})} \leq Me^{-\omega t}, \qquad t \geq 0, \tag{16}$$

for some $M \geq 1$ and $\omega > 0$ independent of λ.

Remark 5 (Finite-Energy Outgoing Covariance Coupling and Finite-Dimensional Calibration). The membership $a_{Ik} \in H_{Q_0}$ is an admissibility condition, not a consequence of the free-block Lyapunov equation alone: an arbitrary outgoing vector need not lie in $\mathrm{Range}((\Sigma_{II}^{(0)})^{1/2})$. It says that the deterministic channel carrying target fluctuations into the free-block has finite covariance energy relative to the free-noise background.

To understand this condition intuitively, consider a finite-dimensional setting where the free-block state space is $\mathcal{H}_I = \mathbb{R}^{d-1}$ and the free stationary covariance $Q_0 \coloneqq \Sigma_{II}^{(0)}$ is strictly positive-definite. Since $Q_0 \succ 0$, its square root $Q_0^{1/2}$ is invertible, and its range is the entire space $\mathbb{R}^{d-1}$. In this case, the Cameron–Martin space is the full space $H_{Q_0} = \mathrm{Range}(Q_0^{1/2}) = \mathbb{R}^{d-1}$, and the Cameron–Martin energy norm $|u|_{Q_0} = \|Q_0^{-1/2}u\|_2$ is equivalent to the standard Euclidean norm. Consequently, the admissibility condition $a_{Ik} \in H_{Q_0}$ is trivially satisfied for any outgoing vector $a_{Ik} \in \mathbb{R}^{d-1}$.

However, if some direction in the free coordinates is noise-free under block decoupling (meaning Q_0 is positive semidefinite but singular), H_{Q_0} becomes a proper subspace of $\mathbb{R}^{d-1}$. In this case, the condition $a_{Ik} \in H_{Q_0}$ requires that the leakage vector a_{Ik} has no component in the null space of Q_0. Equivalently, it prevents the target coordinate from leaking fluctuations into a downstream coordinate that has zero background noise, which would otherwise result in infinite relative covariance energy.

In infinite dimensions, $\Sigma_{II}^{(0)}$ is trace-class and compact, so its eigenvalues decay to zero, making $\mathrm{Range}((\Sigma_{II}^{(0)})^{1/2})$ a dense but strictly proper subspace of $\mathcal{H}_I$. The condition $a_{Ik} \in H_{Q_0}$ requires the off-diagonal drift coupling coefficients to decay sufficiently fast relative to the background noise eigenvalues to keep the relative energy finite.

Under this assumption, invariance and exponential stability on H_{Q_0} imply:

$$
\begin{aligned}
&(\lambda I - \mathcal{A}_{II})^{-1} a_{Ik} \in H_{Q_0}, \\
&|(\lambda I - \mathcal{A}_{II})^{-1} a_{Ik}|_{Q_0} \leq \frac{M}{\lambda + \omega} |a_{Ik}|_{Q_0}.
\end{aligned}
\tag{17}
$$

Remark 6 (Constant Dependency in Schur Energy Expansion). Consequently, the exact leakage cross-covariance $\Sigma_{\lambda,Ik}$ has finite Cameron–Martin energy norm and is $O(\lambda^{-3})$ in that norm. This is a condition on the deterministic response and covariance energy; it does not assert that infinite-dimensional Gaussian sample paths lie in H_{Q_0} almost surely.

3. Inverse-Free Schur Complementation in Hilbert Space

3.1. Variational Shorting and Energy Bounds

In finite dimensions, the Schur complement of the bulk covariance block Σ_{II} is written as $S_k := \Sigma_{kk} - \Sigma_{kI}\Sigma_{II}^{-1}\Sigma_{Ik}$. Under stable and strictly positive diffusion, the bulk covariance Σ_{II} is positive-definite and invertible, so this expression is a valid coordinate formula. Its operator-theoretic content, however, is shorting: for a non-negative block operator, the Schur complement is the quadratic form of the Anderson–Trapp short to the target subspace, and the inverse expression appears only when the complementary block is invertible [5, 6]. In infinite dimensions, the stationary covariance operator is trace-class and compact, so $\Sigma_{\lambda,II}$ lacks a bounded global inverse. The finite-dimensional formula disappears, but the shorted operator remains well-defined through a variational supremum. We therefore formulate the singular-conditioning residual directly in covariance-energy forms.

Let

$$
Q_0 = \int_0^\infty e^{r\mathcal{A}_{II}} D_{II} e^{r\mathcal{A}_{II}^*}\, dr
$$

denote the ambient stable-bulk background covariance from Assumption 4. We define the core Cameron–Martin space $H_{Q_0} := \mathrm{Range}(Q_0^{1/2})$ and the core energy norm:

$$
|u|_{Q_0} := \|Q_0^{-1/2} u\| \quad \text{for all } u \in H_{Q_0}. \tag{18}
$$

By Assumption 4, the off-diagonal drift coupling vector satisfies $a_{Ik} \in H_{Q_0}$, and the restricted drift generator $\mathcal{A}_{II}$ generates an exponentially stable semigroup on H_{Q_0}.

Under this assumption, the exact cross-covariance resolvent formula (derived in Section Appendix A) satisfies:

$$\Sigma_{\lambda,Ik} = \frac{\varepsilon_0}{2\lambda^2}(\lambda I - \mathcal{A}_{II})^{-1} a_{Ik}. \tag{19}$$

This resolvent response yields the Cameron–Martin energy bound on the cross-covariance block:

$$|\Sigma_{\lambda,Ik}|_{Q_0} = O(\lambda^{-3}) \quad \text{as } \lambda \to \infty. \tag{20}$$

To define the infinite-dimensional shorted residual, we introduce the variational Schur subtraction R_λ:

$$R_\lambda \coloneqq \sup_{\langle u, \Sigma_{\lambda,II} u\rangle > 0} \frac{|\langle u, \Sigma_{\lambda,Ik}\rangle|^2}{\langle u, \Sigma_{\lambda,II} u\rangle}. \tag{21}$$

Equivalently, if Σ_λ is viewed as a non-negative block operator on $\mathbb{C}\phi_k \oplus \mathcal{H}_I$, then (21) is the one-dimensional Anderson–Trapp shorted-operator subtraction, namely

$$\left\langle \phi_k, \mathcal{S}_{\mathbb{C}\phi_k}(\Sigma_\lambda)\phi_k \right\rangle = \Sigma_{\lambda,kk} - R_\lambda. \tag{22}$$

The restriction to vectors with $\langle u, \Sigma_{\lambda,II} u\rangle > 0$ simply removes null directions of the denominator; for a non-negative block operator, such directions do not change the supremum. Thus R_λ is the energy removed by shorting to the target coordinate [6, 8, 9, 18].

Since the target noise channel and bulk noise channel are independent, the bulk state decomposes into independent background and target-driven components, meaning the bulk covariance dominates the background covariance ($\Sigma_{\lambda,II} = Q_0 + \mathrm{Cov}(Y_I) \succeq Q_0$). Douglas range inclusion then guarantees that the variational subtraction is bounded:

$$0 \le R_\lambda \le |\Sigma_{\lambda,Ik}|^2_{Q_0} = O(\lambda^{-6}). \tag{23}$$

3.2. Shorted Schur Residual Scaling

The main scaling theorem for the shorted Schur residual follows. The L^2 projection interpretation is secondary and follows from covariance geometry.

Theorem 7 (Hilbert Shorted Schur Invariant)**.** *Under Assumption 4, the L^2-projection residual satisfies:*

$$S_{k,\lambda} \coloneqq \|X_k - P_{\mathcal{M}_I} X_k\|^2_{L^2} = \frac{\varepsilon_0}{2\lambda^2} - R_\lambda, \tag{24}$$

where the remainder satisfies $0 \le R_\lambda = O(\lambda^{-6})$*. In particular, the shorted Schur invariant is:*

$$m_2^{\text{Schur}} = \lim_{\lambda\to\infty} \lambda^2 S_{k,\lambda} = \frac{\varepsilon_0}{2}, \tag{25}$$

and the reciprocal projection-residual scaling satisfies:

$$S_{k,\lambda}^{-1} = \frac{2}{\varepsilon_0}\lambda^2 + O(\lambda^{-2}). \tag{26}$$

This result depends only on covariance and orthogonal projection; it holds for any linear stationary model satisfying these assumptions, without Gaussianity.

Proof. See Section Appendix B for the full derivation. □

3.3. Convergence of Finite-Dimensional Truncations

Let $\Pi_n : \mathcal{H} \to \mathcal{H}_n$ be finite-rank Galerkin projections preserving the target-bulk decomposition. The convergence theorem below is stated for the admissible spectral truncations used in the covariance-energy argument: on the bulk space, Π_n^I truncates in an eigenbasis of Q_0, $Q_0^{(n)} = \Pi_n^I Q_0 \Pi_n^I|_{\Pi_n^I \mathcal{H}_I}$, and $a_{Ik}^{(n)} = \Pi_n^I a_{Ik}$. This choice is not cosmetic; it is what preserves the Cameron–Martin energy uniformly:

$$\sup_n |a_{Ik}^{(n)}|_{Q_0^{(n)}} \le |a_{Ik}|_{Q_0} < \infty. \tag{27}$$

Write $\mathcal{A}_n^{(\lambda)} := \Pi_n \mathcal{A}^{(\lambda)}|_{\mathcal{H}_n}$, $\mathcal{D}_n^{(\lambda)} := \Pi_n \mathcal{D}^{(\lambda)} \Pi_n$, and let $\Sigma_{n,\lambda}$ solve the truncated Lyapunov equation $\mathcal{A}_n^{(\lambda)} \Sigma_{n,\lambda} + \Sigma_{n,\lambda} (\mathcal{A}_n^{(\lambda)})^* + \mathcal{D}_n^{(\lambda)} = 0$.

Proposition 8 (Galerkin Consistency of the Shorted Schur Invariant)**.** *Under Assumption 4, let* $\Pi_n : \mathcal{H} \to \mathcal{H}_n$ *be finite-rank orthogonal Galerkin projections preserving the target-bulk block decomposition:*

$$\Pi_n \phi_k = \phi_k, \qquad \Pi_n \mathcal{H}_I \subset \mathcal{H}_I, \qquad \Pi_n \to I \text{ strongly on } \mathcal{H}. \tag{28}$$

Assume the truncated regularized operator family satisfies the uniform exponential stability and trace-class conditions stated in Lemma 21, and assume either the uniform finite-energy condition

$$\sup_n |a_{Ik}^{(n)}|_{Q_0^{(n)}} < \infty, \tag{29}$$

or, more concretely, the spectral truncation choice $a^{(n)}_{Ik} = \Pi^I_n a_{Ik}$ *and* $Q^{(n)}_0 = \Pi^I_n Q_0 \Pi^I_n$ *in a common* Q_0 *eigenbasis. Then the truncated covariance operators converge in trace norm:*

$$\|\Sigma_{n,\lambda} - \Sigma_\lambda\|_{\mathcal{S}_1} \to 0 \quad \textit{as } n \to \infty, \tag{30}$$

and the truncated Schur invariant is exact at every admissible level:

$$m_2^{\mathrm{Schur},(n)} := \lim_{\lambda\to\infty} \lambda^2 S^{(n)}_{k,\lambda} = \frac{\varepsilon_0}{2} \quad \textit{for all admissible } n. \tag{31}$$

Proof. See Lemmas 21, 22, and 23 for the trace-norm convergence, the spectral finite-energy monotonicity, and the variational shorting support. □

3.4. The Triple Impossibility of Direct Inversion

The variational shorting formula of Section 3.1 is the primitive operator object whose finite-dimensional invertible expression is the Schur formula. The following result is deliberately stated in the GFF spectral regime used by the elliptic example [23, 24], rather than as a general compact-operator theorem.

Theorem 9 (Triple Impossibility in the GFF Spectral Regime)**.** *Let* $Q_0 := \Sigma^{(0)}_{II}$ *be the free-block covariance operator under full decoupling. Assume the GFF spectral regime: in an orthonormal bulk eigenbasis,*

$$Q_0 e_j = \sigma_j e_j, \qquad \sigma_j \asymp j^{-2}, \tag{32}$$

and the corresponding precision or drift eigenvalues satisfy

$$\mu_j \asymp j^2. \tag{33}$$

Let $Q^{(N)}_0$ *be the* N*-mode Galerkin truncation in this basis. Then:*

(i) ***Algebraic impossibility.*** *In infinite dimensions,* Q_0 *is compact and injective with* 0 *in its spectrum. Hence* Q_0^{-1} *is unbounded on the ambient Hilbert space, and the classical Schur expression* $\Sigma_{KK} - \Sigma_{KI}\Sigma^{-1}_{II}\Sigma_{IK}$ *is not an ambient bounded-operator formula.*

(ii) ***Numerical impossibility.*** *The Galerkin covariance matrices have condition numbers*

$$\kappa\big(Q^{(N)}_0\big) = \frac{\max_{1\le j\le N} \sigma_j}{\min_{1\le j\le N} \sigma_j} \asymp N^2. \tag{34}$$

Direct inversion therefore amplifies errors at the same spectral rate as the inverse GFF precision scale.

(iii) ***Asymptotic impossibility.*** *The diagonal resolvent factor* $(\lambda+\mu_j)^{-1}$ *has a critical mode*

$$j^*(\lambda) \asymp \sqrt{\lambda}. \tag{35}$$

Modes $j \ll j^*(\lambda)$ *are in the clamp-dominated regime* $(\lambda+\mu_j)^{-1} \asymp \lambda^{-1}$*, while modes* $j \gg j^*(\lambda)$ *remain in the precision-dominated regime* $(\lambda+\mu_j)^{-1} \asymp \mu_j^{-1} \asymp j^{-2}$*. Hence a fixed truncation dimension* N *cannot uniformly represent the large-*λ *transition unless it resolves the moving scale* $j^*(\lambda)$*.*

The consistent resolution is the variational shorting operator (Section 3.1), the operator object whose invertible-block coordinate expression is the Schur formula. It is well-defined without bounded ambient inverses and its Galerkin approximation is a nested Rayleigh–Ritz restriction of the same variational quantity.

Proof. For (i), compactness of Q_0 follows from $\sigma_j \to 0$. Since the Hilbert space is infinite-dimensional and Q_0 is injective with eigenvalues tending to zero, 0 belongs to the spectrum but is not an eigenvalue. Thus the inverse exists only as an unbounded operator on its range or as a Cameron–Martin form, not as a bounded operator on the ambient Hilbert space.

For (ii), the assumptions $\sigma_j \asymp j^{-2}$ give constants $c, C > 0$ such that $cj^{-2} \le \sigma_j \le Cj^{-2}$. Hence $\max_{1\le j\le N} \sigma_j \asymp \sigma_1 \asymp 1$ and $\min_{1\le j\le N} \sigma_j \asymp \sigma_N \asymp N^{-2}$, proving (34).

For (iii), the transition occurs where λ and μ_j have comparable size. Since $\mu_j \asymp j^2$, the relation $\mu_j \asymp \lambda$ is equivalent to $j \asymp \sqrt{\lambda}$, which proves (35). The two displayed resolvent regimes follow by comparing λ and μ_j on either side of this scale. $\square$

4. Fredholm Trace Anomalies and Gaussian Finite Parts

This section applies the shorted-operator construction to Gaussian Radon laws. Disintegration supplies the exact-evidence measure; shorting supplies the conditional covariance object to which the Gaussian formulas can be applied. The Fredholm finite part is then canonical: the determinant term and the trace-class perturbation generated by the shorted covariance defect cancel at the operator-spectral level.

4.1. Radon Gaussian Application and Disintegration

Assumption 10 (Gaussian Application and Affine Mean Channel)**.** In addition to Assumption 4, let $\mu_\lambda = \mathcal{N}(m_\lambda, \Sigma_\lambda)$ be a Gaussian Radon law on $\mathcal{H}$ with injective trace-class covariance. Let μ_{obs} be the canonical Gaussian disintegration member on the evidence hyperplane $X_k = x_k^*$. Suppose that a fixed deterministic forcing $b \in \mathcal{H}$ gives the stationary mean equation:

$$\mathcal{A}_{\text{cond}} m_\lambda + b - \lambda \Pi_k (m_\lambda - x_k^* \phi_k) = 0, \quad \Pi_k \mathcal{A}_{\text{cond}} = 0. \tag{36}$$

We define the target deterministic offset constant:

$$c_\lambda := \lambda(m_{\lambda,k} - x_k^*) = b_k, \tag{37}$$

where $m_{\lambda,k} := \langle m_\lambda, \phi_k \rangle$ and $b_k := \langle b, \phi_k \rangle$.

Under the Gaussian Radon law, the coordinate projection $\pi_k : \mathcal{H} \to \mathbb{R}$ is continuous, so the target variable X_k is a well-defined Borel random variable. The Polish structure of the Hilbert space guarantees the existence of a regular conditional probability distribution. By selecting the weakly continuous version of the conditional law at the single point $t = x_k^*$ (as detailed in Section Appendix D), we obtain the unique pointwise disintegration member μ_{obs}. Under this measure, the target coordinate satisfies the exact target zeros:

$$m_{\text{obs},k} - x_k^* = 0, \quad \text{Var}_{\text{obs}}(X_k) = 0 \quad \text{(exact)}. \tag{38}$$

4.2. Measurable Cameron–Martin Energy Projection

Let $C_\lambda := \Sigma_{\lambda,II}$ denote the Gaussian bulk covariance operator on $\mathcal{H}_I$. Because C_λ is trace-class and compact, it lacks a bounded global inverse. We define the bulk Cameron–Martin space $H_{C_\lambda} := \text{Range}(C_\lambda^{1/2})$ and the corresponding energy norm.

The leakage cross-covariance vector satisfies $\Sigma_{\lambda,Ik} \in H_{C_\lambda}$ with energy norm $\|\Sigma_{\lambda,Ik}\|_{H_{C_\lambda}} = O(\lambda^{-3})$. This range inclusion allows us to define the energy projection $\mu_{k|I}(X_I) := \mathbb{E}[X_k \mid X_I]$ as a measurable Wiener functional:

$$\mu_{k|I}(X_I) = m_{\lambda,k} + W_{\Sigma_{\lambda,Ik}}(X_I - m_{\lambda,I}), \tag{39}$$

which is the unique linear L^2 predictor. The estimates in Section Appendix D imply that, for all sufficiently large λ, the bulk marginals $\mu_{\text{obs},I}$ and $\mu_{\lambda,I}$ are equivalent and have finite bulk relative entropy:

$$D_{\text{KL}}(\mu_{\text{obs},I} \parallel \mu_{\lambda,I}) = O(\lambda^{-4}) \to 0 \quad \text{as } \lambda \to \infty. \tag{40}$$

4.3. Gaussian Fredholm Finite-Part Functional

Although the joint KL divergence diverges under exact conditioning, the shorted covariance isolates the operator finite part before it is interpreted as a Gaussian density residue.

Definition 11 (Gaussian Fredholm Finite-Part Functional)**.** Whenever $S_{k,\lambda} > 0$ and $\mu_{\mathrm{obs},I}$ is equivalent to $\mu_{\lambda,I}$, we define the Gaussian Fredholm finite-part functional of the conditioned law:

$$\begin{aligned}\mathfrak{J}_\lambda(\mu_{\mathrm{obs}};\mu_\lambda) :=& D_{\mathrm{KL}}(\mu_{\mathrm{obs},I} \parallel \mu_{\lambda,I}) \\ &+ \frac{1}{2} S_{k,\lambda}^{-1} \mathbb{E}_{\mu_{\mathrm{obs}}}\left[\left(X_k - \mu_{k|I}(X_I) \right)^2 \right].\end{aligned} \tag{41}$$

This is the finite part of the joint KL divergence after removing the normalization defect due to the Dirac measure; the remaining covariance contribution is controlled by Fredholm determinant and trace terms.

Weak scheme-independence.. For an evidence level $t \in \mathbb{R}$, write μ_λ^t for the weakly continuous conditional law from Section Appendix D and set

$$\mathfrak{J}_\lambda(t) \coloneqq \mathfrak{J}_\lambda(\mu_\lambda^t;\mu_\lambda).$$

For any two evidence levels t_1, t_2 for which the finite part is defined,

$$\mathfrak{J}_\lambda(t_2) - \mathfrak{J}_\lambda(t_1) = \frac{(t_2 - m_{\lambda,k})^2 - (t_1 - m_{\lambda,k})^2}{2\Sigma_{\lambda,kk}}. \tag{42}$$

In the calibrated scaling $\Sigma_{\lambda,kk} = \varepsilon_0/(2\lambda^2)$, if $b_i = \lambda(m_{\lambda,k} - t_i)$, this becomes

$$\mathfrak{J}_\lambda(t_2) - \mathfrak{J}_\lambda(t_1) = \frac{b_2^2 - b_1^2}{\varepsilon_0}.$$

The full proof is in Appendix Section Appendix D, and it also shows weak-independence from the mollifier: replacing the Gaussian mollifier in Proposition 28 by any centered approximate identity with finite second moment changes the finite part only by a kernel-dependent term independent of t.

4.4. Gaussian Main Theorem

The main Fredholm regularization and cancellation theorems for the finite-part functional follow.

Theorem 12 (Gaussian Fredholm Finite-Part Regularization)**.** *Under Assumption 10, there exists $\lambda_0 > 0$ such that, for $\lambda \geq \lambda_0$, the Gaussian Fredholm finite-part functional under the conditioned law μ_{obs} is well-defined and satisfies:*

$$\mathfrak{J}_\lambda(\mu_{\text{obs}}; \mu_\lambda) = \frac{b_k^2}{\varepsilon_0} + O(\lambda^{-4}). \tag{43}$$

In particular, the functional remains bounded at $O(1)$ and converges to:

$$\lim_{\lambda \to \infty} \mathfrak{J}_\lambda(\mu_{\text{obs}}; \mu_\lambda) = \frac{b_k^2}{\varepsilon_0}. \tag{44}$$

Proof. See Section Appendix D for the full derivation. □

Corollary 13 (Exact Decoupled-Centered Cancellation)**.** *Under Assumption 10, if the target coordinate is decoupled from the bulk ($a_{Ik} = 0$) and the evidence is centered at the stationary mean ($x_k^* = m_{\lambda,k}$), then:*

$$\mathfrak{J}_\lambda(\mu_{\text{obs}}; \mu_\lambda) = 0 \quad \textit{for every } \lambda > 0. \tag{45}$$

Proof. The decoupling implies $\Sigma_{\lambda,Ik} = 0$, so $D_{\text{KL}}(\mu_{\text{obs},I} \parallel \mu_{\lambda,I}) = 0$. The centering makes the conditional residual zero under the conditioning, so both terms in (41) vanish. □

5. Elliptic Gaussian Field Example

A standard elliptic Gaussian field example shows that the abstract Hilbert-space assumptions are natural for covariance operators arising from stochastic evolution equations and elliptic Gaussian priors [3, 12, 22, 23].

5.1. Elliptic Gaussian field on $[0, 1]$

Let

$$L = -\frac{d^2}{dx^2} + \kappa^2$$

on $L^2(0,1)$ with Dirichlet boundary conditions and $\kappa > 0$. Then $\mathcal{A}_0 = -L$ with domain $H^2(0,1) \cap H_0^1(0,1)$ is dissipative, L is positive self-adjoint and sectorial, and e^{-tL} is an exponentially stable analytic contraction semigroup. The inverse covariance

$$Q_0 = L^{-1}$$

is compact, positive, self-adjoint, and trace class. With eigenfunctions $e_j(x) = \sqrt{2}\sin(\pi j x)$, its eigenvalues are

$$q_j = (\pi^2 j^2 + \kappa^2)^{-1}.$$

Thus Q_0 is a typical infinite-dimensional covariance operator: it is compact and trace class, has no bounded inverse on the ambient space, and carries its inverse only as a Cameron–Martin energy form [1, 2, 23].

5.2. Cameron–Martin range and covariance energy

The Cameron–Martin space associated with Q_0 is

$$H_{Q_0} = \mathrm{Range}(Q_0^{1/2}) = H_0^1(0,1),$$

with equivalent energy norm

$$\|h\|_{Q_0}^2 = \langle h, Lh\rangle_{L^2} = \int_0^1 \left(|h'(x)|^2 + \kappa^2 |h(x)|^2\right) dx.$$

This identifies the variational Schur projection in a Sobolev-energy form, as in the standard covariance-energy representation of Gaussian Hilbert priors [1, 12, 23]. The covariance inverse does not act as a bounded operator on $L^2(0,1)$; it acts only through the energy form on its Cameron–Martin domain.

5.3. Spectral sensor conditioning

Fix a target eigenmode e_k and define the observed coordinate

$$X_k = \langle X, e_k\rangle_{L^2}.$$

The unperturbed target variance is

$$\Sigma_{0,kk} = q_k = (\pi^2 k^2 + \kappa^2)^{-1}.$$

A regularized restoring drift on this coordinate produces the exact normalization

$$\Sigma_{\lambda,kk} = \frac{\varepsilon_0}{2\lambda^2}, \qquad S_{k,\lambda} = \frac{\varepsilon_0}{2\lambda^2} - R_\lambda, \qquad R_\lambda = O(\lambda^{-6}),$$

under the block assumptions of Section 3. Off-diagonal bounded perturbations of the drift operator generate the cross-covariance vector entering the variational Schur residual. The resulting subtraction is evaluated by the Cameron–Martin energy form above, not by a global inverse on $L^2(0,1)$.

This example shows that the abstract theorem applies to a genuine Gaussian field. The residual is controlled by a Sobolev energy on $H_0^1(0,1)$ while the ambient covariance remains compact.

6. Conclusion

The paper shows that singular Gaussian conditioning in Hilbert space is not governed by the formal Schur algebra $A - BD^{-1}B^*$. That formula is the finite-dimensional invertible-block shadow of Anderson–Trapp shorting. When the free covariance block is compact and trace class, the shadow disappears but the short remains: the residual is defined by variational shorting on the Cameron–Martin range, and Galerkin convergence follows from nested variational suprema rather than algebraic inversion. The Gaussian conditioning statements then follow as consequences: disintegration supplies the exact-evidence law, and Fredholm determinant/trace identities cancel the finite-rank anomalies, leaving the canonical finite residual. The paper does not treat infinite-rank observation operators or non-Gaussian conditioning theory.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work, the author(s) used Google DeepMind's Antigravity assistant in order to merge the LaTeX manuscripts, clean up cross-referencing tags, format mathematical notation, and edit the draft for style consistency. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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Appendix A. Block Lyapunov Identities for the Calibrated Clamp

This appendix isolates the algebra used in the paper's shorting argument. The only inputs are the block-triangular calibrated operator family, the trace-class stationary covariance, and the Cameron–Martin finite-energy assumption on the outgoing drift channel. All covariance computations below are made for centered variables; deterministic means play no role in the operator identities.

Appendix A.1. Block identities and stationary convolutions

Write $\mathcal{H} = \mathcal{H}_k \oplus \mathcal{H}_I$ with $\mathcal{H}_k = \mathbb{R}\phi_k$. Under the calibrated clamp,

$$\mathcal{A}^{(\lambda)} = \begin{pmatrix} -\lambda & 0 \\ a_{Ik} & \mathcal{A}_{II} \end{pmatrix}, \qquad \mathcal{D}^{(\lambda)} = \begin{pmatrix} \varepsilon_0/\lambda & 0 \\ 0 & D_{II} \end{pmatrix}. \tag{A.1}$$

Let Σ_λ be the stationary covariance solving

$$\mathcal{A}^{(\lambda)}\Sigma_\lambda + \Sigma_\lambda(\mathcal{A}^{(\lambda)})^* + \mathcal{D}^{(\lambda)} = 0.$$

The background bulk covariance is

$$Q_0 = \int_0^\infty e^{r\mathcal{A}_{II}} D_{II} e^{r\mathcal{A}_{II}^*}\, dr, \tag{A.2}$$

which is the stationary covariance of the decoupled bulk equation.

Lemma 14 (Bulk domination by independent stationary convolutions)**.** *The stationary bulk coordinate admits the decomposition*

$$X_I(t) = X_I^{(0)}(t) + Y_{\lambda,I}(t), \qquad Y_{\lambda,I}(t) = \int_{-\infty}^t e^{(t-s)\mathcal{A}_{II}} a_{Ik} X_k(s)\, ds, \tag{A.3}$$

where

$$X_I^{(0)}(t) = \int_{-\infty}^t e^{(t-s)\mathcal{A}_{II}} D_{II}^{1/2}\, dW_I(s). \tag{A.4}$$

The two summands $X_I^{(0)}(t)$ and $Y_{\lambda,I}(t)$ are driven by independent noises and hence have zero cross covariance. Therefore

$$\Sigma_{\lambda,II} = Q_0 + \operatorname{Cov}(Y_{\lambda,I}) \succeq Q_0. \tag{A.5}$$

Proof. The target coordinate is the stationary Ornstein–Uhlenbeck convolution

$$X_k(t) = \int_{-\infty}^t e^{-\lambda(t-s)} \sqrt{\varepsilon_0/\lambda}\, dW_k(s), \tag{A.6}$$

with W_k independent of the bulk cylindrical Brownian motion W_I. Variation of constants for the lower triangular system gives (A.3)–(A.4). The exponential stability assumptions guarantee convergence of the improper stochastic and deterministic convolutions in $L^2(\Omega; \mathcal{H}_I)$.

For any $h, g \in \mathcal{H}_I$, the scalar random variable $\langle h, X_I^{(0)}(t)\rangle$ is measurable with respect to the sigma-field generated by W_I, whereas $\langle g, Y_{\lambda,I}(t)\rangle$ is measurable with respect to the sigma-field generated by W_k. These sigma-fields are independent, and the centered variables therefore have zero covariance. Hence

$$\mathrm{Cov}(X_I(t)) = \mathrm{Cov}(X_I^{(0)}(t)) + \mathrm{Cov}(Y_{\lambda,I}(t)).$$

The first covariance is exactly (A.2). The second is a positive covariance operator, proving the Loewner domination (A.5). $\square$

Lemma 15 (Cross-covariance resolvent identity)**.** *For every $\lambda > 0$,*

$$\Sigma_{\lambda,kk} = \frac{\varepsilon_0}{2\lambda^2}, \tag{A.7}$$

and the target-to-bulk cross covariance, identified with the vector $\Pi_I \Sigma_\lambda \phi_k \in \mathcal{H}_I$, is

$$\Sigma_{\lambda,Ik} = \frac{\varepsilon_0}{2\lambda^2}(\lambda I - \mathcal{A}_{II})^{-1} a_{Ik}. \tag{A.8}$$

The resolvent in (A.8) *exists both on $\mathcal{H}_I$ and, under Assumption 4, on H_{Q_0}. Consequently $\Sigma_{\lambda,Ik} \in H_{Q_0}$.*

Proof. The scalar target Lyapunov equation gives

$$-2\lambda \Sigma_{\lambda,kk} + \frac{\varepsilon_0}{\lambda} = 0,$$

which proves (A.7). Since $\mathcal{A}_{II}$ generates an exponentially stable semigroup, every $\lambda > 0$ belongs to the resolvent set of $\mathcal{A}_{II}$ and

$$(\lambda I - \mathcal{A}_{II})^{-1} x = \int_0^\infty e^{-\lambda r} e^{r\mathcal{A}_{II}} x \, dr, \qquad x \in \mathcal{H}_I. \tag{A.9}$$

The same formula holds on H_{Q_0} for the restricted exponentially stable semigroup from Assumption 4. In particular, $(\lambda I - \mathcal{A}_{II})^{-1} a_{Ik}$ is a well-defined element of H_{Q_0}.

It remains to compute the sign and the scalar coefficient. By Lemma 14, the background bulk convolution is independent of X_k, so only $Y_{\lambda,I}$ contributes to the cross covariance. Using stationarity of the target Ornstein–Uhlenbeck process,

$$\mathbb{E}\big[X_k(t-r)X_k(t)\big] = \Sigma_{\lambda,kk} e^{-\lambda r}, \qquad r \geq 0.$$

Therefore

$$\begin{aligned}\Sigma_{\lambda,Ik} &= \mathrm{Cov}(X_I(t), X_k(t)) \\ &= \int_0^\infty e^{r\mathcal{A}_{II}} a_{Ik} \mathbb{E}\big[X_k(t-r)X_k(t)\big]\, dr \\ &= \Sigma_{\lambda,kk} \int_0^\infty e^{-\lambda r} e^{r\mathcal{A}_{II}} a_{Ik}\, dr \\ &= \frac{\varepsilon_0}{2\lambda^2}(\lambda I - \mathcal{A}_{II})^{-1} a_{Ik}.\end{aligned}$$

Equivalently, the Ik block of the Lyapunov equation is

$$(\mathcal{A}_{II} - \lambda I)\Sigma_{\lambda,Ik} + a_{Ik}\Sigma_{\lambda,kk} = 0,$$

which rearranges to the same positive-sign resolvent formula. □

Appendix A.2. Cameron–Martin energy

Let $Q_0 := \Sigma_{II}^{(0)}$ be the background bulk covariance and $H_{Q_0} := \mathrm{Range}(Q_0^{1/2})$ with energy norm $|u|_{Q_0} = \|Q_0^{-1/2}u\|_{\mathcal{H}_I}$. Assumption 4 states that $a_{Ik} \in H_{Q_0}$ and that the bulk resolvent is bounded on this energy space:

$$|(\lambda I - \mathcal{A}_{II})^{-1}u|_{Q_0} \le \frac{M}{\lambda+\omega}|u|_{Q_0}. \tag{A.10}$$

Lemma 16 (Finite-energy leakage)**.** *The cross covariance belongs to H_{Q_0} and satisfies*

$$|\Sigma_{\lambda,Ik}|_{Q_0} \le \frac{\varepsilon_0 M}{2\lambda^2(\lambda+\omega)}|a_{Ik}|_{Q_0} = O(\lambda^{-3}). \tag{A.11}$$

Proof. Apply (A.10) to the resolvent identity (A.8). This proves both the range statement and the energy bound. □

Lemma 17 (Form-domain energy domination)**.** *Let C and Q be bounded non-negative self-adjoint operators on a Hilbert space and assume $C \succeq Q$. Then*

$$H_Q = \mathrm{Range}(Q^{1/2}) \subseteq \mathrm{Range}(C^{1/2}) = H_C, \tag{A.12}$$

and for every $v \in H_Q$,

$$|v|_C \le |v|_Q. \tag{A.13}$$

In particular, with $C = \Sigma_{\lambda,II}$ and $Q = Q_0$,

$$|\Sigma_{\lambda,II}^{-1/2}v| \le |Q_0^{-1/2}v|, \qquad v \in H_{Q_0}. \tag{A.14}$$

Proof. The Loewner inequality $Q \le C$ is equivalent to $Q^{1/2}(Q^{1/2})^* \le C^{1/2}(C^{1/2})^*$. Douglas' range inclusion theorem therefore gives a contraction K on the ambient Hilbert space such that

$$Q^{1/2} = C^{1/2}K, \qquad \|K\| \le 1. \tag{A.15}$$

This proves (A.12). If $v \in H_Q$, write $v = Q^{1/2}x$ with x chosen as the minimal-norm preimage, so $\|x\| = |v|_Q$. By (A.15), $v = C^{1/2}Kx$. The Cameron–Martin norm $|v|_C$ is the minimal norm among all y with $C^{1/2}y = v$. Hence

$$|v|_C \le \|Kx\| \le \|x\| = |v|_Q.$$

Taking $C = \Sigma_{\lambda,II}$ and $Q = Q_0$ is justified by Lemma 14, and gives (A.14). □

Appendix B. Anderson–Trapp Shorting and the Rank-One Residual

In finite dimensions, the Schur complement is only the bounded-inverse shadow of Anderson–Trapp shorting. The Hilbert space argument therefore starts from shorting and recovers the Schur expression only when the complementary block is boundedly invertible.

Lemma 18 (Schur complement as the bounded-inverse shadow of shorting)**.** *Let $\mathcal{E}$ and $\mathcal{F}$ be Hilbert spaces and let*

$$T = \begin{pmatrix} A & B^* \\ B & D \end{pmatrix} \ge 0$$

act on $\mathcal{E} \oplus \mathcal{F}$. If D is boundedly invertible on $\mathcal{F}$, then the Anderson–Trapp short of T to $\mathcal{E}$ has the quadratic form

$$\langle x, \mathcal{S}_{\mathcal{E}}(T)x\rangle = \langle x, Ax\rangle - \langle Bx, D^{-1}Bx\rangle. \tag{B.1}$$

Thus $B^*D^{-1}B$ is exactly the variational shorting subtraction in the special case where D^{-1} is a bounded operator.

Proof. By the variational definition of the shorted operator,

$$\langle x, \mathcal{S}_{\mathcal{E}}(T)x\rangle = \inf_{y\in\mathcal{F}} \left\langle \begin{pmatrix} x \\ y \end{pmatrix}, T\begin{pmatrix} x \\ y \end{pmatrix} \right\rangle.$$

The displayed quadratic form equals $\langle x, Ax\rangle + 2\,\mathrm{Re}\langle Bx, y\rangle + \langle y, Dy\rangle$. Since D is boundedly invertible and positive, the unique minimizer is $y = -D^{-1}Bx$. Substitution gives (B.1). □

Lemma 19 (Rayleigh quotient representation of covariance energy)**.** *Let C be a bounded non-negative self-adjoint operator on a Hilbert space $\mathcal{F}$ and let $z \in \mathcal{F}$. Then*

$$\sup_{\langle u, Cu\rangle > 0} \frac{|\langle u, z\rangle|^2}{\langle u, Cu\rangle} = \begin{cases} |z|_C^2, & z \in H_C, \\ +\infty, & z \notin H_C, \end{cases} \tag{B.2}$$

where $H_C = \mathrm{Range}(C^{1/2})$ and $|z|_C = \|C^{-1/2} z\|$ is the Cameron–Martin energy norm, interpreted as the minimal preimage norm.

Proof. By the spectral theorem, write $Ce_j = \gamma_j e_j$ on the positive spectral subspace and expand $z = \sum_j z_j e_j + z_0$ with $z_0 \in \ker C$. If $z_0 \neq 0$, then positivity of the denominator is unaffected by adding large multiples of z_0 to u, and the supremum is infinite. Thus finiteness forces $z \perp \ker C$. On the positive spectral subspace,

$$\langle u, Cu\rangle = \sum_j \gamma_j |u_j|^2, \qquad \langle u, z\rangle = \sum_j \overline{u_j} z_j.$$

Cauchy's inequality gives

$$|\langle u, z\rangle|^2 \le \left(\sum_j \gamma_j |u_j|^2\right)\left(\sum_j \frac{|z_j|^2}{\gamma_j}\right).$$

If $\sum_j |z_j|^2/\gamma_j = \infty$, truncating to the first m positive modes shows that the supremum diverges. If this sum is finite, choose $u_j^{(m)} = z_j/\gamma_j$ on the first m positive modes and zero otherwise; the corresponding Rayleigh quotients increase to $\sum_j |z_j|^2/\gamma_j = |z|_C^2$. This proves (B.2). □

For the covariance block Σ_λ on $\mathbb{C}\phi_k \oplus \mathcal{H}_I$, the shorting subtraction is the variational quantity

$$R_\lambda = \sup_{\langle u, \Sigma_{\lambda,II} u\rangle > 0} \frac{|\langle u, \Sigma_{\lambda,Ik}\rangle|^2}{\langle u, \Sigma_{\lambda,II} u\rangle}. \tag{B.3}$$

The null directions of the denominator do not change the supremum for a non-negative block covariance: positivity of the full block forces the cross-vector to annihilate those null directions.

Proposition 20 (Rank-one shorted residual)**.** *Under Assumption 4,*

$$S_{k,\lambda} = \Sigma_{\lambda,kk} - R_\lambda, \qquad R_\lambda = |\Sigma_{\lambda,Ik}|^2_{\Sigma_{\lambda,II}} = \|\Sigma_{\lambda,Ik}\|^2_{H_{\Sigma_{\lambda,II}}}, \tag{B.4}$$

and

$$0 \le R_\lambda \le |\Sigma_{\lambda,Ik}|^2_{Q_0} = O(\lambda^{-6}). \tag{B.5}$$

Consequently

$$S_{k,\lambda} = \frac{\varepsilon_0}{2\lambda^2} - O(\lambda^{-6}), \qquad m_2^{\mathrm{Schur}} = \frac{\varepsilon_0}{2}, \tag{B.6}$$

and

$$S_{k,\lambda}^{-1} = \frac{2}{\varepsilon_0}\lambda^2 + O(\lambda^{-2}).$$

Proof. The first identity $S_{k,\lambda} = \Sigma_{\lambda,kk} - R_\lambda$ is the one-dimensional Anderson–Trapp short of the positive block covariance Σ_λ to the target coordinate. Applying Lemma 19 with $C = \Sigma_{\lambda,II}$ and $z = \Sigma_{\lambda,Ik}$ gives the exact identity

$$R_\lambda = |\Sigma_{\lambda,Ik}|^2_{\Sigma_{\lambda,II}},$$

provided the vector lies in $H_{\Sigma_{\lambda,II}}$.

This range statement follows from the previous section. Namely, Lemma 14 gives $\Sigma_{\lambda,II} \succeq Q_0$, and Lemma 17 gives $H_{Q_0} \subseteq H_{\Sigma_{\lambda,II}}$ with

$$|v|_{\Sigma_{\lambda,II}} \le |v|_{Q_0}, \qquad v \in H_{Q_0}.$$

Since Lemma 16 gives $\Sigma_{\lambda,Ik} \in H_{Q_0}$ and $|\Sigma_{\lambda,Ik}|_{Q_0} = O(\lambda^{-3})$, we obtain

$$0 \le R_\lambda = |\Sigma_{\lambda,Ik}|^2_{\Sigma_{\lambda,II}} \le |\Sigma_{\lambda,Ik}|^2_{Q_0} = O(\lambda^{-6}).$$

Combining this with (A.7) proves (B.6). Finally,

$$S_{k,\lambda} = \frac{\varepsilon_0}{2\lambda^2}\left(1 - \frac{2\lambda^2}{\varepsilon_0}R_\lambda\right), \qquad \frac{2\lambda^2}{\varepsilon_0}R_\lambda = O(\lambda^{-4}),$$

so the reciprocal expansion follows from the scalar Neumann expansion. ☐

Appendix C. Galerkin Approximation and the Failure of Direct Inversion

Galerkin approximation is used here only to approximate trace-class covariances and nested variational shorting problems. It is not a license to replace the Hilbert-space short by a global inverse of a compact covariance block.

Lemma 21 (Trace-norm convergence of Galerkin covariances)**.** *Let* $\Pi_n : \mathcal{H} \to \mathcal{H}_n$ *be finite-rank orthogonal projections with* $\Pi_n \phi_k = \phi_k$, $\Pi_n \mathcal{H}_I \subset \mathcal{H}_I$, *and* $\Pi_n \to I$ *strongly. For fixed* $\lambda > 0$, *set*

$$\mathcal{A}_n^{(\lambda)} = \Pi_n \mathcal{A}^{(\lambda)}|_{\mathcal{H}_n}, \qquad \mathcal{D}_n^{(\lambda)} = \Pi_n \mathcal{D}^{(\lambda)} \Pi_n.$$

Assume that $e^{t\mathcal{A}_n^{(\lambda)}} x \to e^{t\mathcal{A}^{(\lambda)}} x$ *for every* x, *uniformly for* t *in compact intervals, that the semigroups are uniformly exponentially stable, satisfying*

$$\|e^{t\mathcal{A}_n^{(\lambda)}}\|, \ \|e^{t\mathcal{A}^{(\lambda)}}\| \le M e^{-\beta t},$$

and that $\mathcal{D}_n^{(\lambda)} \to \mathcal{D}^{(\lambda)}$ *in* $\mathcal{S}_1(\mathcal{H})$. *Then*

$$\Sigma_{n,\lambda} = \int_0^\infty e^{t\mathcal{A}_n^{(\lambda)}} \mathcal{D}_n^{(\lambda)} e^{t\mathcal{A}_n^{(\lambda)*}} \, dt \longrightarrow \Sigma_\lambda \quad \textit{in } \mathcal{S}_1(\mathcal{H}).$$

Proof. Omitting the superscript (λ), write the difference as a Bochner integral and split the integrand:

$$\begin{aligned}\|e^{t\mathcal{A}_n} \mathcal{D}_n e^{t\mathcal{A}_n^*} - e^{t\mathcal{A}} \mathcal{D} e^{t\mathcal{A}^*}\|_{\mathcal{S}_1} \le M^2 e^{-2\beta t} &\|\mathcal{D}_n - \mathcal{D}\|_{\mathcal{S}_1} \\ &+ \|e^{t\mathcal{A}_n} \mathcal{D} e^{t\mathcal{A}_n^*} - e^{t\mathcal{A}} \mathcal{D} e^{t\mathcal{A}^*}\|_{\mathcal{S}_1}.\end{aligned}$$

The first term tends to zero. For the second, Grümm's theorem for trace ideals implies trace-norm convergence from strong convergence of the left and right semigroups, uniform boundedness, and $\mathcal{D} \in \mathcal{S}_1$ [2, 3]. The full integrand is dominated by

$$M^2 e^{-2\beta t} \big(\|\mathcal{D}_n\|_{\mathcal{S}_1} + \|\mathcal{D}\|_{\mathcal{S}_1} \big),$$

with a finite uniform trace-norm bound because $\mathcal{D}_n \to \mathcal{D}$ in $\mathcal{S}_1$. Dominated convergence for Bochner integrals gives the claim. □

Lemma 22 (Spectral Galerkin finite-energy monotonicity)**.** *Assume that the Galerkin projections are spectral truncations of the background bulk covariance. Thus, on* $\mathcal{H}_I$,

$$Q_0 e_j = q_j e_j, \qquad q_j > 0, \qquad \Pi_n^I u = \sum_{j=1}^n \langle u, e_j \rangle e_j.$$

Let $Q_0^{(n)} = \Pi_n^I Q_0 \Pi_n^I|_{\Pi_n^I \mathcal{H}_I}$ *and* $a_{Ik}^{(n)} = \Pi_n^I a_{Ik}$. *If* $a_{Ik} \in H_{Q_0}$, *then*

$$|a_{Ik}^{(n)}|^2_{Q_0^{(n)}} = \sum_{j=1}^{n} \frac{|\langle a_{Ik}, e_j\rangle|^2}{q_j} \uparrow |a_{Ik}|^2_{Q_0}. \tag{C.1}$$

In particular,

$$\sup_n |a_{Ik}^{(n)}|_{Q_0^{(n)}} \le |a_{Ik}|_{Q_0} < \infty. \tag{C.2}$$

Proof. Writing $a_j = \langle a_{Ik}, e_j\rangle$, the Cameron–Martin norm of the full vector is

$$|a_{Ik}|^2_{Q_0} = \sum_{j=1}^{\infty} \frac{|a_j|^2}{q_j} < \infty.$$

The truncated covariance has eigenvalues $q_1, \dots, q_n$ on $\Pi_n^I \mathcal{H}_I$, and $a_{Ik}^{(n)} = \sum_{j=1}^n a_j e_j$. Therefore (C.1) follows directly from the spectral formula for the finite-dimensional Cameron–Martin norm. The partial sums are monotone and bounded by the full sum, giving (C.2). □

Lemma 23 (Galerkin convergence of variational shorting)**.** *For every block-preserving Galerkin truncation satisfying the hypotheses of Lemma 21 and the uniform finite-energy condition*

$$a_{Ik}^{(n)} \in H_{Q_0^{(n)}}, \qquad \sup_n |a_{Ik}^{(n)}|_{Q_0^{(n)}} < \infty,$$

the truncated residual satisfies

$$S_{k,\lambda}^{(n)} = \frac{\varepsilon_0}{2\lambda^2} - R_{n,\lambda}, \qquad 0 \le R_{n,\lambda} \le C\lambda^{-6},$$

with C *independent of* n. *Hence* $m_2^{\mathrm{Schur},(n)} = \varepsilon_0/2$ *for every admissible truncation. Moreover, for fixed* λ, *nested finite-dimensional test spaces in the bulk Cameron–Martin space give monotone Rayleigh–Ritz convergence of the variational subtraction to* R_λ.

Proof. The block-preserving truncation has the same triangular form as (A.1), so the finite-dimensional Lyapunov blocks give

$$\Sigma_{n,\lambda,kk} = \frac{\varepsilon_0}{2\lambda^2}, \qquad \Sigma_{n,\lambda,Ik} = \frac{\varepsilon_0}{2\lambda^2}(\lambda I - \mathcal{A}_{II}^{(n)})^{-1} a_{Ik}^{(n)}.$$

The uniform finite-energy assumption and the uniform $H_{Q_0^{(n)}}$ resolvent bound give $|\Sigma_{n,\lambda,Ik}|_{Q_0^{(n)}} = O(\lambda^{-3})$ uniformly in n. The same Douglas domination argument used in Proposition 20 then yields $0 \le R_{n,\lambda} \le C\lambda^{-6}$ and the stated value of $m_2^{\mathrm{Schur},(n)}$. For spectral truncations $a_{Ik}^{(n)} = \Pi_n^I a_{Ik}$, the required uniform finite-energy condition is provided by Lemma 22.

For the variational convergence statement, let $C_\lambda = \Sigma_{\lambda,II}$ and let V_m be nested finite-dimensional subspaces whose union is dense in $H_{C_\lambda} = \mathrm{Range}(C_\lambda^{1/2})$. Then

$$R_\lambda^{[m]} = \sup_{\substack{u \in V_m \\ \langle u, C_\lambda u\rangle > 0}} \frac{|\langle u, \Sigma_{\lambda,Ik}\rangle|^2}{\langle u, C_\lambda u\rangle}$$

is monotone increasing in m and bounded above by R_λ. Density in the Cameron–Martin energy norm gives $R_\lambda^{[m]} \uparrow R_\lambda$. In an eigenbasis of C_λ, this is the Rayleigh–Ritz identity

$$R_\lambda^{[m]} = \sum_{j=1}^{m} \frac{|\langle \Sigma_{\lambda,Ik}, e_j\rangle|^2}{\sigma_j} \uparrow \|\Sigma_{\lambda,Ik}\|_{H_{C_\lambda}}^2.$$

□

The obstruction to direct inversion is not numerical bookkeeping. In the GFF spectral regime it can be stated as a precise spectral theorem.

Lemma 24 (GFF spectral obstruction to direct inversion)**.** *Assume that the background covariance and the bulk drift are simultaneously represented in a bulk eigenbasis $(e_j)_{j\ge1}$ such that*

$$Q_0 e_j = \sigma_j e_j, \qquad \sigma_j \asymp j^{-2},$$

and the positive damping eigenvalues of $-\mathcal{A}_{II}$ satisfy

$$\mu_j \asymp j^2.$$

Let $Q_0^{(N)}$ be the N-mode Galerkin truncation in this basis. Then:

(i) Q_0 is compact and injective with 0 in its spectrum. Hence Q_0^{-1} is not a bounded ambient Hilbert-space operator, and a formula containing the global inverse of the bulk covariance block is not an operator identity on the ambient space.

(ii)

$$\kappa(Q_0^{(N)}) = \frac{\max_{1\le j\le N}\sigma_j}{\min_{1\le j\le N}\sigma_j} \asymp N^2.$$

Thus direct Galerkin inversion amplifies errors at the inverse GFF precision scale.

(iii) The diagonal resolvent filter $(\lambda+\mu_j)^{-1}$ has a moving critical mode $j^*(\lambda)$ determined by $\mu_{j^*(\lambda)} \asymp \lambda$, and hence

$$j^*(\lambda) \asymp \sqrt{\lambda}.$$

For $j \ll j^*(\lambda)$ one has $(\lambda+\mu_j)^{-1} \asymp \lambda^{-1}$, whereas for $j \gg j^*(\lambda)$ one has $(\lambda+\mu_j)^{-1} \asymp \mu_j^{-1} \asymp j^{-2}$. Therefore a fixed Galerkin dimension N does not resolve the large-λ transition unless N grows on the moving scale $\sqrt{\lambda}$.

Consequently, the infinite-dimensional object that survives the singular limit is the Anderson–Trapp variational short, not the direct inverse of a compact bulk covariance block.

Proof. The first claim follows from the spectral theorem: $\sigma_j > 0$ and $\sigma_j \to 0$, so Q_0 is injective and compact, while 0 is a spectral point. The inverse is defined only on its range as an unbounded operator, equivalently as the Cameron–Martin energy form.

For the condition number, the two-sided estimate $\sigma_j \asymp j^{-2}$ gives $\max_{1\le j\le N}\sigma_j \asymp \sigma_1 \asymp 1$ and $\min_{1\le j\le N}\sigma_j \asymp \sigma_N \asymp N^{-2}$, proving $\kappa(Q_0^{(N)}) \asymp N^2$.

For the critical mode, the transition of $(\lambda+\mu_j)^{-1}$ occurs when the two summands in the denominator are comparable. Since $\mu_j \asymp j^2$, the relation $\mu_j \asymp \lambda$ is equivalent to $j \asymp \sqrt{\lambda}$. The two resolvent regimes follow by comparing μ_j with λ below and above this scale. □

Appendix D. Gaussian Radon Conditioning and the Fredholm Finite Part

The Gaussian and Fredholm layer is the Radon representation of the shorted residual constructed above. It is not a second mechanism: disintegration supplies the exact-evidence law, while the shorted covariance residual supplies the finite conditional variance used by the Gaussian entropy calculation.

Appendix D.1. Pointwise disintegration

Let $\mu_\lambda = \mathcal{N}(m_\lambda, \Sigma_\lambda)$ be a Radon Gaussian measure on $\mathcal{H} = \mathcal{H}_k \oplus \mathcal{H}_I$, and let $\pi_k(X) = \langle X, \phi_k \rangle$. Write

$$\sigma_\lambda = \Sigma_{\lambda,kk}, \qquad C_\lambda = \Sigma_{\lambda,II}, \qquad z_\lambda = \Sigma_{\lambda,Ik}.$$

Lemma 25 (Gaussian disintegration covariance)**.** *For each $t \in \mathbb{R}$, there is a unique weakly continuous Gaussian disintegration member $\mu_\lambda^t = \mathcal{N}(m_\lambda^t, \Sigma_\lambda^t)$ supported on $X_k = t$, where*

$$m_\lambda^t = \begin{pmatrix} t \\ m_{\lambda,I} + z_\lambda \sigma_\lambda^{-1}(t - m_{\lambda,k}) \end{pmatrix},$$

and

$$\Sigma_\lambda^t = \begin{pmatrix} 0 & 0 \\ 0 & C_\lambda - z_\lambda \sigma_\lambda^{-1} z_\lambda^* \end{pmatrix}. \tag{D.1}$$

The bulk conditional covariance

$$C_\lambda^t := C_\lambda - z_\lambda \sigma_\lambda^{-1} z_\lambda^* \tag{D.2}$$

is a positive trace-class covariance operator on $\mathcal{H}_I$. At $t = x_k^*$, the observed law $\mu_{\text{obs}} = \mu_\lambda^{x_k^*}$ satisfies

$$\mathbb{E}_{\text{obs}}[X_k] - x_k^* = 0, \qquad \mathrm{Var}_{\text{obs}}(X_k) = 0.$$

Proof. The map π_k is a continuous one-dimensional linear map and its marginal variance is $\sigma_\lambda > 0$. Regular conditional probabilities therefore exist because $\mathcal{H}$ is Polish. To identify the continuous Gaussian version, center the variables and set

$$\widetilde{X}_I = X_I - m_{\lambda,I} - z_\lambda \sigma_\lambda^{-1}(X_k - m_{\lambda,k}).$$

For every $h \in \mathcal{H}_I$,

$$\mathrm{Cov}\big(\langle h, \widetilde{X}_I \rangle, X_k - m_{\lambda,k}\big) = \langle h, z_\lambda \rangle - \langle h, z_\lambda \rangle \sigma_\lambda^{-1} \sigma_\lambda = 0.$$

The pair is jointly Gaussian, hence $\widetilde{X}_I$ is independent of X_k. Conditioning on $X_k = t$ therefore shifts only the deterministic affine part and leaves the covariance of $\widetilde{X}_I$ unchanged. Direct covariance calculation gives

$$\mathrm{Cov}(\widetilde{X}_I) = C_\lambda - z_\lambda \sigma_\lambda^{-1} z_\lambda^*.$$

This proves the displayed conditional mean and covariance. The operator C^t_λ is positive because it is a covariance. It is trace-class because C_λ is trace-class and $z_\lambda \sigma_\lambda^{-1} z_\lambda^*$ is rank one. The affine formulas are continuous in t, so this gives the unique weakly continuous disintegration member along the scalar observation map [1, 19, 20, 25]. Evaluating at x_k^* fixes the target coordinate almost surely. □

Appendix D.2. Gaussian energy projection and entropy

Since C_λ is trace-class, its inverse is not a bounded ambient operator; the relevant object is the Cameron–Martin space $H_{C_\lambda} = \mathrm{Range}(C_\lambda^{1/2})$.

Lemma 26 (Measurable Cameron–Martin energy projection)**.** *The cross covariance belongs to H_{C_λ} and*

$$\|z_\lambda\|_{H_{C_\lambda}} \le |z_\lambda|_{Q_0} = O(\lambda^{-3}).$$

The conditional mean projection is the measurable Wiener functional

$$\mu_{k|I}(X_I) = m_{\lambda,k} + W_{z_\lambda}(X_I - m_{\lambda,I}),$$

and, for all sufficiently large λ,

$$\mathrm{Var}(X_k \mid X_I) = \sigma_\lambda - \|z_\lambda\|^2_{H_{C_\lambda}} = S_{k,\lambda} > 0.$$

Proof. The domination $C_\lambda = \Sigma_{\lambda,II} \succeq Q_0$ and Lemma 17 give $H_{Q_0} \subseteq H_{C_\lambda}$ with contraction of energy norms. Combining this inclusion with (A.11) gives the norm bound. The range condition $z_\lambda \in H_{C_\lambda}$ is exactly the condition for the measurable Wiener functional associated with z_λ to exist under the bulk Gaussian law [1, 19]. The variance identity is the one-dimensional shorting identity from Proposition 20; positivity holds for large λ because $S_{k,\lambda} = \varepsilon_0/(2\lambda^2) - O(\lambda^{-6})$. □

Lemma 27 (Feldman–Hajek rank-one perturbation)**.** *There is $\lambda_0 > 0$ such that for $\lambda \ge \lambda_0$ the bulk marginals $\mu_{\mathrm{obs},I}$ and $\mu_{\lambda,I}$ are equivalent. The relative covariance perturbation has at most one nontrivial eigenvalue, namely*

$$1 - \rho_\lambda, \qquad \rho_\lambda = \frac{\|z_\lambda\|^2_{H_{C_\lambda}}}{\sigma_\lambda} = 1 - \frac{S_{k,\lambda}}{\sigma_\lambda} = O(\lambda^{-4}). \tag{D.3}$$

Moreover, if $\Delta_\lambda = x_k^* - m_{\lambda,k}$, then

$$2D_{\mathrm{KL}}(\mu_{\mathrm{obs},I} \parallel \mu_{\lambda,I}) = \frac{\Delta_\lambda^2}{\sigma_\lambda}\rho_\lambda - \log(1-\rho_\lambda) - \rho_\lambda. \tag{D.4}$$

In particular, $D_{\mathrm{KL}}(\mu_{\mathrm{obs},I} \parallel \mu_{\lambda,I}) = O(\lambda^{-4})$.

Proof. Put $g_\lambda = C_\lambda^{-1/2} z_\lambda$ and $\alpha_\lambda = \|g_\lambda\|^2 = \|z_\lambda\|^2_{H_{C_\lambda}}$. If $z_\lambda = 0$, then $\rho_\lambda = 0$ and the covariance statement is trivial. Otherwise let $e_\lambda = g_\lambda/\|g_\lambda\|$. The conditional bulk covariance from (D.2) can be written in relative form as

$$C_\lambda^{x_k^*} = C_\lambda^{1/2}(I - \rho_\lambda e_\lambda \otimes e_\lambda)C_\lambda^{1/2}, \qquad \rho_\lambda = \alpha_\lambda/\sigma_\lambda. \tag{D.5}$$

Thus the relative covariance is the identity on $e_\lambda^\perp$ and has the single nontrivial eigenvalue $1 - \rho_\lambda$ along e_λ.

By Proposition 20, $\alpha_\lambda = R_\lambda = O(\lambda^{-6})$, while $\sigma_\lambda = \varepsilon_0/(2\lambda^2)$. Hence $\rho_\lambda = O(\lambda^{-4})$ and, for all sufficiently large λ, $0 \le \rho_\lambda < 1$. Then $I - \rho_\lambda e_\lambda \otimes e_\lambda$ is boundedly positive and invertible, so the Cameron–Martin ranges of $C_\lambda^{x_k^*}$ and C_λ coincide. The covariance perturbation is rank one, hence Hilbert–Schmidt in relative coordinates. The mean shift is $z_\lambda \sigma_\lambda^{-1} \Delta_\lambda \in H_{C_\lambda}$ and has C_λ-energy squared

$$\left\| C_\lambda^{-1/2} z_\lambda \sigma_\lambda^{-1} \Delta_\lambda \right\|^2 = \frac{\Delta_\lambda^2}{\sigma_\lambda} \rho_\lambda.$$

Feldman–Hajek therefore gives equivalence of the two bulk Gaussian laws; recent Hilbert-space Gaussian inverse-problem work uses the same measure-class condition for finite-rank posterior updates [13, 14].

The Gaussian relative entropy formula for equivalent Gaussian measures gives one contribution from the mean and one from the nontrivial relative covariance eigenvalue:

$$2D_{\mathrm{KL}} = \frac{\Delta_\lambda^2}{\sigma_\lambda}\rho_\lambda + \left[(1-\rho_\lambda) - 1 - \log(1-\rho_\lambda)\right],$$

which is (D.4); see also recent Hilbert-space log-determinant/KL treatments [13, 21]. Since $\Delta_\lambda = O(\lambda^{-1})$, $\sigma_\lambda = O(\lambda^{-2})$, and $\rho_\lambda = O(\lambda^{-4})$, the mean contribution is $O(\lambda^{-4})$; the logarithmic remainder is $O(\rho_\lambda^2)$ after cancellation of the linear term. Thus the bulk entropy is $O(\lambda^{-4})$. □

Appendix D.3. Mollified divergence and finite part

For $\delta > 0$, set

$$\nu_{\lambda,\delta} = \mu_{\mathrm{obs},I} \otimes \mathcal{N}(x_k^*, \delta^2)$$

under the product identification $\mathcal{H} = \mathcal{H}_I \times \mathcal{H}_k$.

Proposition 28 (Mollification limit). *For $\lambda \ge \lambda_0$ as in Lemma 27, define*

$$C_{\lambda,\delta} = \frac{1}{2}\left(\log\frac{S_{k,\lambda}}{\delta^2} - 1\right).$$

Then

$$\lim_{\delta\downarrow 0}\left[D_{\mathrm{KL}}(\nu_{\lambda,\delta}\parallel\mu_\lambda)-C_{\lambda,\delta}\right]=\mathfrak{J}_\lambda(\mu_{\mathrm{obs}};\mu_\lambda).$$

Proof. Let $r_\lambda(X)=X_k-\mu_{k|I}(X_I)$. The entropy chain rule and the scalar Gaussian formula give

$$D_{\mathrm{KL}}(\nu_{\lambda,\delta}\parallel\mu_\lambda)=D_{\mathrm{KL}}(\mu_{\mathrm{obs},I}\parallel\mu_{\lambda,I})+C_{\lambda,\delta}+\frac{\delta^2}{2S_{k,\lambda}}+\frac{1}{2S_{k,\lambda}}\mathbb{E}_{\mu_{\mathrm{obs}}}[r_\lambda(X)^2].$$

The δ^2 term vanishes as $\delta\downarrow 0$, and the remaining terms are exactly Definition 11. □

Lemma 29 (KL finite-part cancellation algebra)**.** *Let $t\in\mathbb{R}$, let μ_λ^t be the Gaussian disintegration member, and set $\Delta_t=t-m_{\lambda,k}$. For $\lambda\geq\lambda_0$,*

$$D_{\mathrm{KL}}(\mu_{\lambda,I}^t\parallel\mu_{\lambda,I})=\frac{1}{2}\frac{\Delta_t^2}{\sigma_\lambda}\rho_\lambda-\frac{1}{2}\log(1-\rho_\lambda)-\frac{1}{2}\rho_\lambda,\tag{D.6}$$

and

$$\frac{1}{2S_{k,\lambda}}\mathbb{E}_{\mu_\lambda^t}\left[(X_k-\mu_{k|I}(X_I))^2\right]=\frac{1}{2}\rho_\lambda+\frac{1}{2}\frac{\Delta_t^2}{\sigma_\lambda}(1-\rho_\lambda).\tag{D.7}$$

Consequently,

$$\mathfrak{J}_\lambda(t)=\frac{\Delta_t^2}{2\sigma_\lambda}-\frac{1}{2}\log(1-\rho_\lambda).\tag{D.8}$$

Proof. Equation (D.6) is (D.4) with x_k^* replaced by t; this is the rank-one specialization of the standard equivalent-Gaussian KL formula [1, 21]. It remains to compute the second term. Under μ_λ^t, the target coordinate is fixed at $X_k=t$. Let

$$Z_\lambda=W_{z_\lambda}(X_I-m_{\lambda,I}).$$

The conditional bulk mean shift is $z_\lambda\sigma_\lambda^{-1}\Delta_t$; therefore

$$\mathbb{E}_{\mu_\lambda^t}Z_\lambda=\langle z_\lambda,z_\lambda\sigma_\lambda^{-1}\Delta_t\rangle_{H_{C_\lambda}}=\rho_\lambda\Delta_t.$$

The conditional bulk covariance is $C_\lambda-z_\lambda\sigma_\lambda^{-1}z_\lambda^*$, so

$$\mathrm{Var}_{\mu_\lambda^t}(Z_\lambda)=\|z_\lambda\|_{H_{C_\lambda}}^2-\sigma_\lambda^{-1}\|z_\lambda\|_{H_{C_\lambda}}^4=\sigma_\lambda\rho_\lambda(1-\rho_\lambda).$$

Since $X_k-\mu_{k|I}(X_I)=\Delta_t-Z_\lambda$ under μ_λ^t,

$$\mathbb{E}_{\mu_\lambda^t}\left[(X_k-\mu_{k|I}(X_I))^2\right]=\sigma_\lambda\rho_\lambda(1-\rho_\lambda)+\Delta_t^2(1-\rho_\lambda)^2.$$

Dividing by $2S_{k,\lambda} = 2\sigma_\lambda(1-\rho_\lambda)$ gives (D.7). Adding (D.6) and (D.7) cancels the $\rho_\lambda/2$ terms and also combines the two signal factors:

$$\frac{1}{2}\frac{\Delta_t^2}{\sigma_\lambda}\rho_\lambda + \frac{1}{2}\frac{\Delta_t^2}{\sigma_\lambda}(1-\rho_\lambda) = \frac{\Delta_t^2}{2\sigma_\lambda}.$$

This proves (D.8). □

Lemma 30 (Weak scheme-independence)**.** *For any two evidence levels t_1, t_2 for which the finite part is defined,*

$$\mathfrak{J}_\lambda(t_2) - \mathfrak{J}_\lambda(t_1) = \frac{(t_2 - m_{\lambda,k})^2 - (t_1 - m_{\lambda,k})^2}{2\Sigma_{\lambda,kk}}. \tag{D.9}$$

Replacing the Gaussian Dirac mollifier in Proposition 28 by any centered approximate identity with finite second moment changes the extracted finite part only by an additive term depending on the kernel, λ, and $S_{k,\lambda}$, not on t.

Proof. The exact identity (D.8) shows that the only t-dependent term is $\Delta_t^2/(2\sigma_\lambda)$. The logarithmic term $-\frac{1}{2}\log(1-\rho_\lambda)$ depends on C_λ and λ, but not on t, and therefore cancels in the difference. The same chain-rule computation as in Proposition 28 shows that changing the centered mollifier changes only the one-dimensional normalization and centered second moment contribution, which are independent of the evidence level t. □

Proof of Theorem 12. Apply Lemma 29 with $t = x_k^*$ and $\Delta_\lambda = x_k^* - m_{\lambda,k}$. Assumption 10 gives $m_{\lambda,k} - x_k^* = b_k/\lambda$, hence $\Delta_\lambda^2 = b_k^2/\lambda^2$. Since $\sigma_\lambda = \varepsilon_0/(2\lambda^2)$,

$$\frac{\Delta_\lambda^2}{2\sigma_\lambda} = \frac{b_k^2}{\varepsilon_0}.$$

By Lemma 27, $\rho_\lambda = O(\lambda^{-4})$, and therefore

$$-\frac{1}{2}\log(1-\rho_\lambda) = O(\lambda^{-4}).$$

The exact finite-part identity (D.8) gives

$$\mathfrak{J}_\lambda(\mu_{\text{obs}};\mu_\lambda) = \frac{b_k^2}{\varepsilon_0} + O(\lambda^{-4}).$$

□

If $a_{Ik} = 0$ and $x_k^* = m_{\lambda,k}$, then $z_\lambda = 0$, the bulk marginals coincide, and the conditional residual term in Definition 11 is zero. Hence $\mathfrak{J}_\lambda(\mu_{\text{obs}};\mu_\lambda) = 0$ exactly for every $\lambda > 0$.

Appendix E. Admissibility Boundary and Elliptic Benchmark

The finite-energy assumption is a boundary condition for the theory, not a technical artifact. The off-diagonal channel must land in the Cameron–Martin range of the background covariance. If it does not, the variational subtraction need not be finite and the $O(\lambda^{-6})$ rank-one residual estimate has no meaning.

The elliptic example of Section 5 makes this boundary concrete. For $L = -d^2/dx^2 + \kappa^2$ on $L^2(0,1)$ with Dirichlet boundary conditions and $Q_0 = L^{-1}$, the Cameron–Martin space is

$$H_{Q_0} = H_0^1(0,1), \qquad \|h\|_{Q_0}^2 = \int_0^1 \left(|h'(x)|^2 + \kappa^2 |h(x)|^2\right) dx.$$

Smooth spatially averaged observations are admissible when their representing vectors lie in $H_0^1(0,1)$; for example, any fixed $g \in C_c^\infty(0,1)$ has finite energy. A point-evaluation channel is the opposite limit. If g_ε is a standard compactly supported mollifier concentrating at $x_0 \in (0,1)$, then

$$\|g_\varepsilon\|_{Q_0}^2 = \int_0^1 \left(|g_\varepsilon'(x)|^2 + \kappa^2 |g_\varepsilon(x)|^2\right) dx \longrightarrow \infty \qquad (\varepsilon \downarrow 0).$$

Thus each fixed smoothed sensor can satisfy the finite-energy condition, but the Dirac point-evaluation limit does not. The shorting theorem is therefore a theory of admissible Cameron–Martin channels, not of unsmoothed point sensors outside the covariance-energy domain.